Weight and local gravity

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish mass from gravitational weight.

SUCCESS CRITERIA - I can...

1. Use mass in kg and force in N.
2. Calculate weight using the stated local g.
3. Explain how weight changes while mass stays the same.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “weight” and “mass” to explain your answer. Sentence starters are optional.

weight: the gravitational force on an object

mass: a measure of how strongly an object resists acceleration

field: a description of an influence at each location

01 Warm-up: bring back an earlier idea

A **5 kg** cart starts from rest under **35 N** net force right for **4 s**.
1. **Find acceleration** and final velocity.
2. **State one thing you** are treating as true.

Word help: net force = the vector sum of all forces on one object

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_g = mg$; $g = F_g/m$.
F_g is weight. Field strength in N/kg is numerically equal to free-fall acceleration in m/s².

Worked reasoning: A **3 kg** object in a field of **4 N/kg** has weight **3×4=12 N**. Moving it to a **2 N/kg** field halves weight to **6 N** but leaves mass **3 kg**.

Your turn: Draw weight arrows for the two places and **label the unchanged quantity**.

Sentence starter: In my work, I used … to show … . I labeled it … .

Location	Mass (kg)	g (N/kg)	Weight (N)
A	3	4	12
B	3	2	6

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The same **2 kg** object visits worlds with g=**10** and g=**5 N/kg**. **Compare its mass** and weight.

Sentence starter: Both … . They differ in … because … .

World	g (N/kg)
A	10
B	5

Marzano check - circle: ① ② ③

04 Apply the physics

On World B, g = **5 N/kg**. **Complete the equipment table**. **Explain why doubling the packed** mass doubles its weight.

Sentence starter: I think … . I can point to …, which shows …, so … .

Equipment mass (kg)	Weight (N)
3	____
6	____

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A scale reports **0 N** for an astronaut in an orbiting spacecraft. Does that prove Earth exerts no gravitational force? Explain.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** object has **32 N** right and **8 N** left. **Draw the forces** and **find net force** and acceleration.
Draw the forces on the chosen object and **label their sources**.

Word help: acceleration = rate of change of velocity

Sentence starter: I found … [unit]. This value means … in this situation.

Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

A **4 kg** object is where g=**3 N/kg**. **State its mass** and weight.

Sentence starter: I notice … . This supports … because … .

Add and label force arrows.

Draw and label the gravitational force.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two objects weigh the same on different worlds. Must their masses be the same? Build a numerical counterexample.

Sentence starter: Both … . They differ in … because … .

Case	Mass (kg)	g (N/kg)	Weight (N)
1	____	____	____
2	____	____	____

Marzano check - circle: 1 2 3